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Design of magnetized liner inertial fusion experiments using the Z facility^{a)}

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The magnetized liner inertial fusion concept has been presented as a path toward obtaining substantial thermonuclear fusion yields using the Z accelerator [S. A. Slutz *et al.*, Phys. Plasmas **17**, 056303 (2010)]. We present the first integrated magnetohydrodynamic simulations of the inertial fusion targets, which self-consistently include laser preheating of the fuel, the presence of electrodes, and end loss effects. These numerical simulations provided the design for the first thermonuclear fusion neutron-producing experiments on Z using capabilities that presently exist: peak currents of $I_{max} = 18\text{--}20$ MA, pre-seeded axial magnetic fields of $B_z^0 = 10$ T, laser preheat energies of about $E_{las} = 2$ kJ delivered in 2 ns, *DD* fuel, and an aspect ratio 6 solid *Be* liner imploded to 70 km/s. Specific design details and observables for both near-term and future experiments are discussed, including sensitivity to laser timing and absorbed preheat energy. The initial experiments measured stagnation radii $r_{stag} < 75\text{ }\mu\text{m}$, temperatures around 3 keV, and isotropic neutron yields up to $Y_n^{DD} = 2 \times 10^{12}$, with inferred alpha-particle magnetization parameters around $r_{stag}/r_L^\alpha = 1.7$ [M. R. Gomez *et al.*, Phys. Rev. Lett. (submitted)]. © 2014 AIP Publishing LLC.
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I. INTRODUCTION

Pulsed-power accelerators, such as the Z facility,^{1,2} generate large MA-scale currents, and thereby produce strong magnetic fields at small radius, in order to achieve Mbar pressures in materials. Direct magnetically-driven implosions for inertial confinement fusion (ICF) are an interesting subclass of magneto-inertial fusion (also called magnetized target fusion) approaches^{3,4} because the absorbed target energies and coupling efficiencies are both significantly enhanced⁵ compared with indirect radiation-driven ICF approaches.^{5–8} The MagLIF (*Magnetized Liner Inertial Fusion*) concept has been presented as a path toward economically obtaining substantial thermonuclear fusion yields on pulsed-power accelerators via the implosion of cylindrical metal liners containing pre-magnetized and laser-preheated fuel.⁹ The first neutron-producing experiments of the concept have occurred with encouraging preliminary results.¹⁰

Over the past several decades, numerous ICF groups have studied the roles of magnetic fields in inertial fusion targets because of their effect on heat transport and the mobility of *DT* fusion-produced He^4 particles, also called alpha particles. To name a few examples, experiments conducted with an electron-beam-driven target at Sandia National Laboratories demonstrated enhanced fusion yield with fuel magnetization.¹¹ Subsequent calculations investigated the parameter regime of magnetized ICF targets driven by lasers or charged-particle beams.^{12–14} Z-pinch gas puff experiments at the University of California demonstrated significant flux compression of an embedded axial magnetic field relevant for enhanced fusion.¹⁵ Scientists in Germany and Russia have

examined heavy-ion-beam-driven approaches to the cylindrical implosion of pre-magnetized metallic tubes containing preheated fuel.^{16–18} Researchers involved in magnetized target fusion experiments at the Los Alamos National Laboratory and Air Force Research Laboratory generated preheated plasmas confined in a closed-field-line magnetic topology called the field-reversed configuration, which were intended to be translated into and compressed by a magnetically-driven liner implosion system on the μs timescale.^{19–22} The Laboratory for Laser Energetics performed radiation-driven implosions of spherical capsules, and diagnosed higher hot-spot temperatures, magnetic flux compression, and enhanced fusion yields when the target was pre-magnetized.^{23–25} A team at the Lawrence Livermore National Laboratory computationally investigated pre-magnetization of indirect-drive ignition targets for the National Ignition Facility and found the imposed field may improve capsule stability and hot-spot formation, while relaxing the conditions needed for ignition.²⁶ Ultimately, each proposed fusion approach seeks to harness magnetic fields in order to reduce electron thermal conduction losses from the heated plasma, enhance particle and energy confinement, and lower the stagnation pressures required to reach fusion conditions, and therefore enable the use of drivers with lower energy and power.

The MagLIF concept utilizes magnetic pressure in a direct-drive approach to compress the fusion fuel, and related experiments have begun in earnest at Sandia National Laboratories. Initial simulations⁹ indicated *DT* fuel that was axially pre-magnetized, laser-preheated, and then imploded on an accelerator similar to the Z machine (peak currents $I_{max} \sim 20\text{--}70$ MA in 100–300 ns) could potentially be compressed to achieve Gbar pressures and high fusion yields (fusion energies $E_{fus} \sim 0.1\text{--}100$ MJ). The target load to be imploded consists of a solid cylindrical metal liner, composed

^{a)}Paper C12 1, Bull. Am. Phys. Soc. **58**, 60 (2013).

^{b)}Invited speaker.

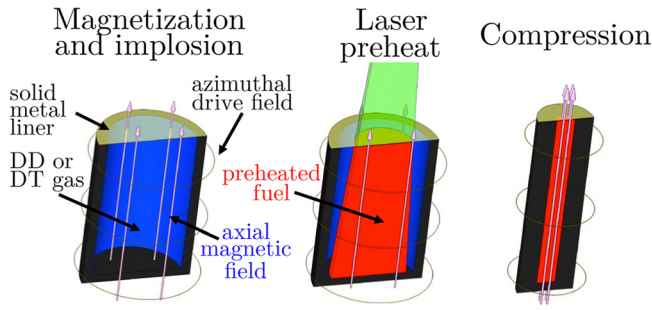


FIG. 1. Schematic of the MagLIF concept.

of pure *Be* or *Al* and about 1 cm in height, containing a high-pressure gas fill of fuel (gas densities $\rho_g \sim 1\text{--}5\text{ mg/cc}$). A schematic of the target is provided in Fig. 1. First, an axial magnetic field ($B_z^0 \sim 5\text{--}100\text{ T}$) is generated throughout the load region on the μs timescale by external capacitors and magnetic coils positioned near the load. Second, when the pulsed-power accelerator discharges, an azimuthal B_θ magnetic field is formed around the liner by the large MA-level axial currents, and via the $J_z \times B_\theta$ Lorentz force, begins to drive the liner implosion, which occurs on the 10s of ns timescale. Third, as the liner begins to implode, laser energy ($E_{\text{las}} \sim 2\text{--}25\text{ kJ}$) is coupled to the fuel through a laser entrance hole (LEH) in the top of the target, causing the fuel to be preheated (ion temperatures $T_i \sim 50\text{--}500\text{ eV}$). The resulting order $\sim 1\text{ Mbar}$ highly conductive plasma has high β (the ratio of thermal to magnetic pressure) and the B_z field is partially frozen into the plasma. The quasi-adiabatic implosion is subsonic and the liner accelerates until peak implosion velocity v_{imp} just prior to deceleration and stagnation at small radius on the plasma pressure near the axis. The resulting areal density ρR of the fuel is low by typical ICF standards ($\sim 1\text{--}10\text{ mg cm}^{-2}$). Since instability and asymmetry can prevent reaching high implosion stagnation pressures P_{stag} , the required convergence ratio CR (defined as the ratio of initial gas radius to stagnated radius) should be minimized.

Volumetric compression in cylindrical geometry is reduced compared with spherical. However, by preheating the fuel before its compression and by reducing its radial thermal electron conduction losses via magnetothermal insulation, lower liner-imparted PdV work to the fuel may be used to achieve fusion-relevant $T_i > 4\text{ keV}$ temperatures with $CR \approx 20\text{--}30$ and $v_{\text{imp}} \leq 100\text{ km/s}$ (compared with $CR > 30$ and $v_{\text{imp}} > 300\text{ km/s}$ in traditional ICF). This is a consequence of the fact that fuel magnetization in low ρR fuel enables reduced driver power, which scales as $\sim (\rho R)^2$ for fixed ignition temperature in cylindrical targets.²⁷ The initial B_z^0 is flux-compressed to $B_z^f \geq 5\text{--}10\text{ kT}$ ($\geq 50\text{--}100\text{ MG}$) at stagnation, and the typical ignition threshold for the fuel ρR ($\sim 0.3\text{--}1\text{ g cm}^{-2}$) is replaced with one for $B_z^f R$,²⁷ or equivalently R/r_L^z , where r_L^z is the Larmor radius ($m_\alpha v_\perp / qB_z$) of the fusion alpha particles. Inertial confinement for the volumetric fusion burn is provided by the liner ρR instead. Magnetic Reynolds numbers vary from a few 10^3 to 10^4 , and the Hall parameter $\omega_{ce}\tau_e$, the product of electron cyclotron frequency and electron-ion collision time, can exceed a few hundred at its peak. Due to magnetization of the alpha particles

produced in the hot spot, they can provide self-heating of the thermonuclear plasma. The P_{stag} required to achieve fusion conditions in this approach is lowered to a few Gbar, from the 100s of Gbar necessary in traditional ICF.

Magnetically-driven MagLIF implosions are substantially different than radiation-driven ICF implosions. As already mentioned, many traditional performance metrics are significantly eased (v_{imp} , dv/dt , ρR , and P_{stag}). Since pulsed power is energy-rich, the liners are thick and massive with aspect ratios of order ~ 10 (ratio of outer radius to liner thickness) in order to improve implosion robustness to magneto-Rayleigh-Taylor (MRT) instability.²⁸ Radiography data from liner implosions on Z prior to integrated MagLIF experiments suggest the inner liner surface is sufficiently stable up to at least $CR \approx 7$,^{29–32} and the presence of the B_z field may provide further stabilization.³¹ The liner materials contain no high atomic number dopants normally used to prevent x-ray preheat of the fuel, drive symmetry is not influenced by the presence of complex radiation flux from laser-plasma interaction, and simulations suggest the fuel can withstand higher amounts of mix.⁹ Due to the drive pressure dependence $P_{\text{dr}} \sim I^2 r_l^{-2}$, where r_l is the outer radius of the liner, peak velocity and acceleration occur at the end of the implosion (when the in-flight aspect ratio is order unity), implying improved stability. Hot spot formation is externally-supplied via laser preheating, not from spherically-converging well-timed shocks. At stagnation, the liner is not ablated away but becomes a compressed tamper with high ρR . For these reasons, the realization of high convergence with few ns confinement times at stagnation is possible, and the first integrated MagLIF experiments¹⁰ suggest $CR \sim 40$ was achieved.

Focused experiments involving each of the necessary components for MagLIF are being conducted using the Z facility and Z-Beamlet (ZBL) laser.³³ The focused experiments study the issues of current delivery, liner stability, flux compression, and laser heating of pre-magnetized gas. Current delivery is presently limited to $I_{\text{max}} \sim 18\text{--}20\text{ MA}$ by the higher-than-usual inductance of the load power feed in use to accommodate the magnets. Evidence has not been found for shorting within the transmission lines due to the applied $B_z \hat{z}$ crossing the gap, in agreement with 3D particle simulations, which also predict enhanced insulation at late times due to the B_z field. Initial simulations using 27 MA suggested aspect ratio 6 liners made of *Be* may be sufficiently stable to instability growth during the implosion.⁹ Liner stability and flux compression experiments using only imploding liners and B_z field (no laser energy) have occurred at reduced current,^{31,32} wherein inner surface liner integrity appeared to be maintained throughout the implosion and probes measured the change in fringe-field strength above the imploding region, respectively. Laser-only experiments of pre-magnetized targets are being conducted with ZBL, using a laser wavelength $\lambda_{\text{las}} = 527\text{ nm}$ (frequency-doubled 2ω light from Nd:glass) in order to infer energy absorption and penetration depth into the gas. Simulation benchmarking efforts to these focused experiments are occurring simultaneously with integrated MagLIF design and interpretation of experiments, but will be reported in future publications.

This paper provides the details of the computational integrated design effort that led to the first successful thermonuclear fusion neutron-producing MagLIF experiments.¹⁰ The simulations self-consistently include laser preheating of the fuel, the presence of electrodes, and end loss effects, and represent pre-shot optimized performance expectations both for near-term experiments with present capabilities as well as future experiments with upgraded capabilities. Detailed comparison between the experimental observables and post-shot simulations will be provided in a subsequent publication. This paper is organized as follows. The computational model is presented in Sec. II along with an ideal 1D simulation of the near-term target. Section III discusses the details included in integrated modeling of targets as they are shot on Z. Peak performance expectations using capabilities available in near-term experiments are reported in Sec. IV. Section V provides results for an upgraded experiment that could occur on Z, as well as a high-gain target on a future facility. A short summary and discussion are given in Sec. VI.

II. COMPUTATIONAL MODEL AND 1D MagLIF SIMULATION

A. The MagLIF target design model

HYDRA is a massively-parallel, multi-physics, 3D radiation-hydrodynamics design code that is routinely used to simulate inertial confinement fusion experiments.³⁴ The HYDRA model used herein includes arbitrary Lagrangian-Eulerian (ALE) mesh control, 3D laser ray tracing, 3D resistive magnetohydrodynamics (MHD) with general circuit models, flux-limited multi-group local thermal and radiation diffusion, thermonuclear reactions and multi-group diffusion for the resulting charged particles, and local thermal equilibrium (LTE) opacities and equations of state (EOS) for the materials. New MHD boundary conditions allow the use of both the imploding B_θ magnetic field as well as the imposed B_z field to be compressed, and the conductivity and fusion burn models have been updated to include anisotropic transport effects due to the magnetic field. Thermoelectric effects, such as inclusion of the Nernst term, which redistributes magnetic flux in a magnetized plasma due to $\nabla \times (B \times \nabla T)/(eB\omega_{ce}\tau_e)$, are not yet included in the code; the term is less important for weaker gradients and larger magnetic fields. The laser absorption model accounts for inverse bremsstrahlung and ponderomotive effects, but laser plasma interaction effects, such as backscattered light from stimulated Raman and Brillouin scattering, are not included. Nonlocal transport corrections due to long mean free path effects are likewise not included. Future modeling improvements will include the Nernst term and could use advanced models, such as implicit Monte Carlo radiation transport, Monte Carlo transport of burn products, and non-LTE opacities and EOS.

B. Simulation of the near-term experiment in 1D

Ideal 1D HYDRA simulations of near-term MagLIF experiments provide best-case performance estimates against which more realistic integrated calculations can be compared. The point design in the original work⁹ used DT fuel,

$I_{max} = 27$ MA (95 kV charging voltage), $B_z^0 = 30$ T, and $E_{las} = 13$ kJ cm⁻¹, but cannot be immediately realized on Z. The capabilities presently available for experiments are DD fuel, $I_{max} = 18$ –20 MA (80 kV charging voltage), $B_z^0 = 10$ T, and laser energies of about $E_{las} = 2$ kJ delivered in 2 ns by ZBL. The liner parameters are changed to match the implosion time using the reduced drive. The solid Be liner has aspect ratio 6, outer radius $r_l = 0.279$ cm, mass 138 mg cm⁻¹, and contains fuel gas density of $\rho_g = 1.5$ mg/cc. Note the ρ_g is well below the 15 and 18 mg/cc critical densities for laser propagation ($n_{crit} \approx 4.4 \times 10^{21}$ cm⁻³ for $\lambda_{las} = 527$ nm [2 ω] light) in DD and DT, respectively. Experiments will initially use an imploding liner length of 7.5 mm. The assumed liner length is important because the circuit model³⁵ accounts for the load inductance and the fixed deposition energy determines the preheated plasma temperature T_0 prior to compression; this leads to monotonically decreasing yields with increasing liner lengths in 1D, but neglects the important effects of laser coupling and end losses.

As the liner begins to implode at $t = 80$ ns due to direct magnetic pressure, the simulation increments the internal energy of the gas to deposit an equivalent 2 kJ into the electrons over 2 ns within the laser radius in order to mock up the laser preheat not included in 1D calculations. Using an 80 kV charge voltage on Z, the peak current reached during the implosion is $I_{max} = 19$ MA by 120 ns and the liner achieves an implosion velocity $v_{imp} \approx 70$ km/s just prior to stagnation, as shown in Fig. 2. The gas and liner absorbed energies are $E_g^{stag} \approx 50$ kJ cm⁻¹ and $E_l^{stag} \approx 550$ kJ cm⁻¹, respectively. The 1D convergence ratio at stagnation is $CR_{1D} \approx 28$ due to the low level of preheat energy available; the liner converges until pressure balance with the gas is reached, therefore reduced preheat energies increase the CR. By stagnation and peak burn around 140 ns, the fuel assembly shown in Fig. 2 has radius $r_g^{stag} \approx 84$ μ m, $\rho_g \approx 0.6$ g/cc, $\rho R_g \approx 6$ mg cm⁻², and peak $T_i \approx 5$ keV, resulting in about 2.5 Gbar of pressure. The stagnated plasma's ρ , B_z , and T_i are approximately 60% of the values predicted by purely adiabatic compression, which does not account for conduction, radiation, or flux loss. The liner compression has achieved $\rho_l \approx 70$ g/cc and $\rho R_l \approx 1.0$ g cm⁻².

The compressed B_z^f flux in the fuel reaches 40–75 MG, although about $\Phi_{loss} = 36\%$ of the initial flux has been lost.

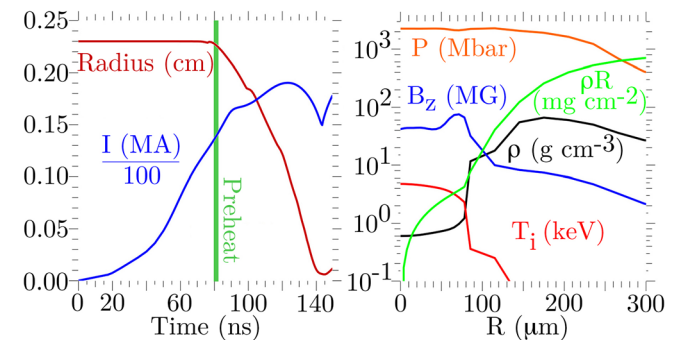


FIG. 2. Simulation of near-term MagLIF experiment in 1D. On the left, liner trajectory (red) and current (blue). On the right, stagnation profiles at $t = 140$ ns of ρ in g/cc (black), T_i in keV (red), ρR in mg cm⁻² (green), B_z in MG (blue), and P in Mbar (orange).

The preheating phase causes an abrupt jump in flux loss (about 20%), but the loss from the gas to the liner grows in time during the rest of the implosion. Radiative heating of the inside liner layer creates the shelf profile in density near the gas-liner boundary, and conductivity details there control the level of flux loss. The magnetized fuel parameter $r_{stag}/r_L^\alpha = 1.5$ at stagnation, or equivalently $B_z^f r_{stag} = 4.1 \times 10^5$ G cm, and falls short of the magnetized ignition criterion estimated in Ref. 27.

The fusion package tracks the three branches of relevant reactions, $D + D \rightarrow He^3$ (0.82 MeV) + n (2.45 MeV), $D + D \rightarrow T$ (1.01 MeV) + p (3.02 MeV), and $D + T \rightarrow He^4$ (3.52 MeV) + n (14.1 MeV), where the product T particle from the initially pure DD fuel can subsequently fuse and generate He^4 alpha particles and secondary neutrons. The primary (2.45 MeV) fusion neutron yield is $Y_n^{DD} = 3.4 \times 10^{14}$ cm⁻¹, or $E_{fus} = 0.4$ kJ cm⁻¹, over a burn duration $t_{burn} \approx 6$ ns ($t_{burn}^{FWHM} = 3.2$ ns). The secondary (14.1 MeV) fusion neutron yield from product T particles fusing with reactant D is $Y_n^{DT} = 1.5 \times 10^{13}$ cm⁻¹, a factor of 23 lower than the primary yield. The energy spectrum and yields of the primary and secondary neutrons are measured at the Z facility using time-of-flight methods and time-integrated activation samples.³⁶

A thermonuclear origin of the neutrons implies an isotropic yield would be measured by probes located in separate radial and axial directions. In the absence of nonthermal fuel ions and when the burn duration is sufficiently short, the spectral width of the primary neutrons provides an estimate of the burn-weighted average ion temperature $\langle T_i \rangle$, which is $\langle T_i \rangle = 2.9$ keV in the above simulation and lower than the 5 keV peak. Additionally, the fuel ρR may be inferred in high- ρR plasma from the ratio of down-scattered neutrons to unscattered neutrons,^{8,37} or in low- ρR plasma from the primary to secondary yield ratio $Y_n^{ratio} = Y_n^{DD}/Y_n^{DT}$ produced by an initially-pure DD fuel.^{38,39} According to Ref. 39, $Y_n^{ratio} \approx 1500$ is expected for this low- ρR stagnation plasma, however the simulation reports a value of 23. Recalling that $r_{stag}/r_L^\alpha = 1.5$, the obvious missing physics is the magnetic field, which increases the effective path length of the product T particle in the fuel since its Larmor radius is about equal to that of the alpha particle. Enhanced confinement of the T due to the high $B_z^f r_{stag}$ value, rather than the ρR , provides for the slowing in the plasma and extra secondary fusion events that lead to larger Y_n^{DT} and smaller Y_n^{ratio} values than implied by Ref. 39, and so the ratio becomes a measure of magnetization and alpha-trapping instead.

The simulated fusion yields assume no mix of $Z > 1$ atomic number elements within the fuel. The thermonuclear production process is sensitive to temperature loss due to the presence of high- Z radiators, and small amounts of such mix can strongly suppress neutron yields.³⁶ The MagLIF target has a number of potential sources of mix, which will be discussed in Sec. III, but the potential elements include Be (from the liner itself), C , N , and O (from the polyimide window), and Al , Cu , or other components of the electrodes. Noble gasses like Ar and Kr are typical in ICF experiments for their diagnostic value to x-ray spectroscopy.

In order to assess yield degradation due to the presence of contaminant elements mixed into the burning plasma, the

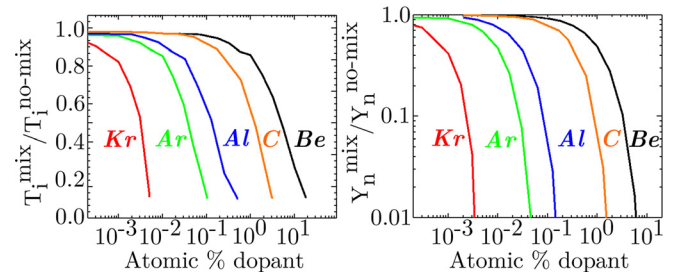


FIG. 3. Normalized degradation of ion temperature (left) and neutron yield (right) in near-term MagLIF targets due to mix of $Z > 1$ elements into the fuel.

1D calculation of the near-term experiment is repeated with various concentrations of Be , C , Al , Ar , and Kr , in order of increasing Z , premixed volumetrically into the fuel. The results from those simulations are shown in Fig. 3 in the form of ion temperature reductions at stagnation and the subsequent decline in neutron yields. Evidently, MagLIF performance is heavily influenced by radiative losses and the yield may be halved by those five elements at atomic percentage concentrations of 1%, 0.3%, 0.025%, 0.009%, and 0.0007%, or reduced by an order of magnitude by 3%, 0.9%, 0.08%, 0.025%, and 0.0025%, respectively. In order to estimate the performance of targets in the presence of laser-ablation-produced mix, integrated calculations are performed.

III. DESCRIPTION OF AN INTEGRATED MODEL FOR MagLIF EXPERIMENTS

An integrated model seeks to realistically simulate experiments as they would occur on the Z accelerator. The parameters and constraints of the target design are self-consistently included and integrated into a single simulation in order to account for details not included in idealized 1D models. An integrated simulation initialization is shown in Fig. 4. The model includes the laser deposition from the top, the laser entrance hole (LEH) and its thin window, and the electrodes at both ends of the liner, allowing for evaluation and optimization of component interactions. Issues to investigate include laser timing and deposition, evolution of the preheated plasma and the entrained B_z field, instability growth on the outer surface of the liner, fuel mass loss and flux loss during the implosion, boundary effects from the electrodes and LEH, and the assembly and evolution of the burning plasma.

Laser parameters include total energy, pulse shape, and spot size. The goal is to heat the fuel T_i to an optimal value specific to the design's initial flux B_z^0 , implosion velocity v_{imp} , and convergence ratio (CR). The focal plane of ZBL's $f/10$ beam is nominally 6–20 mm above the LEH so the desired spot size is incident upon the window and the beam defocuses into the gas. Some energy is required to disassemble the window to below the critical density so the laser can penetrate into the gas. Unlike 1D simulations that deposit energy within the gas at a prescribed rate, integrated modeling evolves the plasma temperatures and magnetic flux consistently using the asymmetric one-sided preheat. The laser can refract during window interaction, launch shocks, generate vorticity, and filament during its propagation.

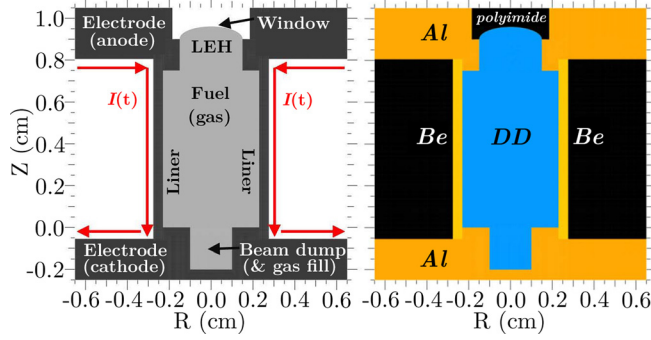


FIG. 4. Integrated MagLIF initialization.

The laser entrance hole is specified by its diameter in the anode and the window thickness and location. The LEH must be large enough to allow deposition into the gas and avoid ablation of anode material (from energy in the laser edge and pointing error) that could refract the beam, introduce mix into the fuel, or prematurely close the hole, while also being as small as possible to minimize loss of the heated fuel during compression. The window thickness scales linearly with both gas density and LEH diameter and is chosen to only be as thick as required to hold the gas pressure (≤ 30 atm) in order to minimize the laser energy it absorbs. Polyimide is used and the initial deformation and stretching from gas pressure is included in the model. The window is recessed from the main fuel region to prevent disassembled window material from mixing with fuel to be compressed by the liner.

A circuit model³⁵ generates the current to implode the liner in the presence of the electrodes. The liner parameters include material, mass, aspect ratio, length, and initial surface roughness. The liner length is important from the perspective of end losses and axially trapping the alpha particles at stagnation, but cannot be arbitrarily long because the preheat energy is fixed and the inductance reduces the current delivery. Electrodes provide electrical contact and boundaries for the target, and their presence allows for effects between the walls and the liner (wall instability), the fuel (cooling), the LEH and beam dump (flow losses), the B_z (bent field lines), and the laser (direct and radiative ablation).

The laser preheating occurs at a chosen time relative to the implosion. An optimal laser beam radius r_{las} exists for a given window thickness, gas density, and liner length. For the most efficient preheating in a given design, the r_{las} is found by maximizing the absorbed energy into the gas without directly ablating either of the electrodes or the inner surface of the liner, since contaminant material may mix into the fuel from just an estimated few hundred J of deposited energy. The gas density choice influences the window thickness, the laser absorption depth for a given r_{las} , and the equilibration temperature of the plasma. Since the pressure generated by a laser is $P_{las} \sim (I_{las}/\lambda_{las})^{2/3}$,⁶ where I_{las} is the laser irradiance in W cm^{-2} , beams with smaller radii generate higher pressures, penetrate the window faster, and lose less energy to window disassembly. However, the resulting inverse bremsstrahlung heating in the fuel is hotter and the laser light penetrates the gas more rapidly,⁹ and may directly ablate cathode material. Window and electrode material motions are influenced by interaction

with the high-pressure plasma. The chosen initial B_z^0 affects laser deposition since larger fields result in hotter and more collimated plasma, and thus increased absorption depths due to lower opacity. Loss of fuel compression efficiency can occur due to B_z flux loss from the fuel (increased thermal conduction loss) as well as hydrodynamic flow loss out the ends of the target. These losses of fuel energy and temperature due to wall and end effects increase the required CR compared with 1D, reduce the yield, and increase experimental risk.

A multi-block mesh is used in the HYDRA code for integrated MagLIF modeling. Blocks can be flexibly arranged in physical and logical space to allow fine resolution on all material interfaces through the use of multiple enhanced and reduced points of connectivity. Aside from flexibility and efficiency advantages, the technique also avoids the potentially problematic effects of nonuniform zoning in unnecessary locations, such as one would have in a single-block mesh (one set of simply-connected physical and logical indices) that attempts to allow fine resolution everywhere it is needed.

The most realistic simulations are three-dimensional. Such r - θ - z simulations can evaluate the effects of instability growth from azimuthally-uncorrelated surface roughness, non-axisymmetric drive and implosion, and laser beam off-centeredness and non-axisymmetric preheating. However, running all design variations in full 3D is impractical, so most calculations assume axisymmetry for expediency. Absent a high level of liner instability growth which may disrupt its ability to uniformly compress the fuel, the effect of most concern missing in 2D r - z simulations is nonuniform heating by the laser.

Insight into non-axisymmetric laser preheating can be gained via r - θ simulation using a spatially-nonuniform absorption pattern representative of an off-centered and unsmoothed laser beam containing hot spots and irregular structure. This case uses $\rho_g = 3 \text{ mg/cc}$, $B_z^0 = 30 \text{ T}$, $r_l = 0.32 \text{ cm}$, and the stagnation time is $t_{stag} = 145 \text{ ns}$, similar to the point design in Ref. 9. A highly nonuniform profile containing hot spots deposits 13 kJ cm^{-1} into the electrons over the period $t = 70$ – 75 ns . Profiles of T_i from the simulation are shown in Fig. 5. The blast wave first encounters the closest inner liner boundary; so long as the inner boundary is not strongly perturbed, axisymmetric compression can still occur. Although significant temperature asymmetry exists initially, collisional equilibration of the electrons and ions occurs 20–25 ns after the heating phase. The plasma

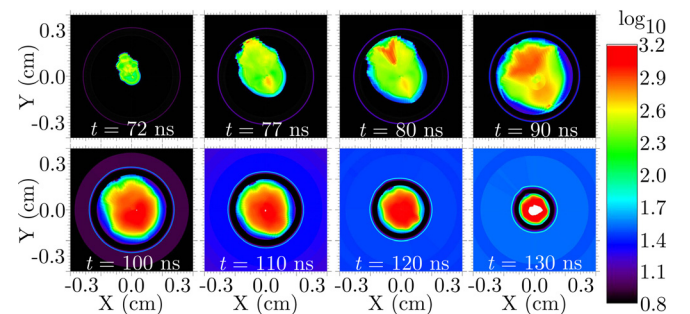


FIG. 5. Nonuniform preheat r - θ simulation. T_i in eV (shown) equilibrates with T_e 20–25 ns after the heating phase and recovers approximate axisymmetry by $CR_{2D} \approx 2$ during the implosion, about 40–45 ns after the heating phase.

temperature nearly recovers axisymmetry before the implosion achieves $CR_{2D}=2$, about 40–45 ns after the heating phase, since several sound transit times have elapsed over this period (typical preheated plasma sound speeds are $C_s \approx 200\text{--}400\text{ km/s}$). Hence, assuming axisymmetry is not an unjustified approximation because the non-uniformly heated plasma is predicted to isotropize during a typical $\sim 50\text{--}70\text{ ns}$ implosion. The simulations throughout the remainder of this paper are executed in 2D r - z geometry.

IV. DESIGN OF NEAR-TERM EXPERIMENTS

A. Design for the first integrated MagLIF target

The initialization for the first experiments is as shown in Fig. 4 and $B_z^0 = 10\text{ T}$. The Be liner has aspect ratio 6, imploding region liner length of $L_l = 7.5\text{ mm}$, contains $\rho_g = 1.5\text{ mg/cc}$ of DD fuel, and carries the $I_{max} = 19\text{ MA}$ current pulse provided in Fig. 2. For initial experiments, the LEH diameter is fixed at 3 mm as a balance between accommodating the laser beam and flow losses of the heated fuel. The thickness at the apex of the elliptically-deformed window is $dz_{win} \approx 2.0\text{ }\mu\text{m}$.

The laser energy of $E_{las} = 2.2\text{ kJ}$ is an $f/10$ beam with a 0.4-TW, 0.5-ns pre-pulse included 2.5 ns prior to the 1-TW, 2.0-ns main pulse ($I_{las} \approx 1.5 \times 10^{14}\text{ W cm}^{-2}$). Both pulses have 0.3 ns rise- and fall-times. The radial beam profile is defined to be an $n = 4$ supergaussian. The use of a pre-pulse reduces the energy invested into window disassembly, and hence increases absorption efficiency into the gas, because it allows time between the pulses for hydrodynamic motion to lower the density. The current and liner trajectory are similar to those shown in Fig. 2. The time $t = 80\text{ ns}$ is chosen to fire the laser and corresponds to when the liner begins imploding, as discussed in Sec. IV C; the preheating ends at $t = 85.6\text{ ns}$.

The optimal preheating laser radius for the L_l , ρ_g , and dz_{win} in this configuration is $r_{las} = 460\text{ }\mu\text{m}$ at the window. The gas absorbs $E_g^{abs} \approx 1.74\text{ kJ}$, of which 1.53 kJ is internal energy, whereas the window absorbs $E_{win}^{abs} \approx 370\text{ J}$.

Profiles of density, electron and ion temperature, and B field magnitude are given in Fig. 6 at several times. The main implosion phase for the liner takes about 60 ns. The laser deposition and B flux evolution can be complicated by filamentary structures and vorticity generation, and a magnetic cavity is produced with flux pushed up against the liner boundary for a short time. Rebounding plasma motion and interacting shocks advect the flux back towards the origin, and axial symmetry is approximately achieved later in time regardless of the one-sided heating. Loss of fuel mass and flux ensues from the end of the heating phase and persists throughout the implosion.

Approximately 25–30 ns after the laser pulse ends, and while the liner is still imploding, the plasma equilibrates to an average $T \approx 250\text{ eV}$ before significant PdV work is done to it by the liner. About 10 ns before stagnation, the central portion of the fuel reaches 1 keV. At stagnation and peak burn around $t = 140\text{ ns}$, the convergence ratio of the liner at the midplane increases to $CR_{2D} \approx 37$ from $CR_{1D} \approx 28$. Generally, integrated $CR_{2D} \geq CR_{1D}$ because of imperfect laser coupling efficiency, fuel mass loss, and potentially degraded B_z flux compression, so the liner must converge farther to reach pressure equilibrium and stagnation. If the liner implosion is sufficiently 3D, symmetry and stability effects will limit the attained CR and stagnation pressure.

An interesting feature can be seen at the ends of the stagnation column where the liner, gas, and electrodes meet. Two effects in tandem cause the liner convergence to be higher at the ends compared with the midplane. Loss of plasma occurs at the ends, leading to lower gas pressure

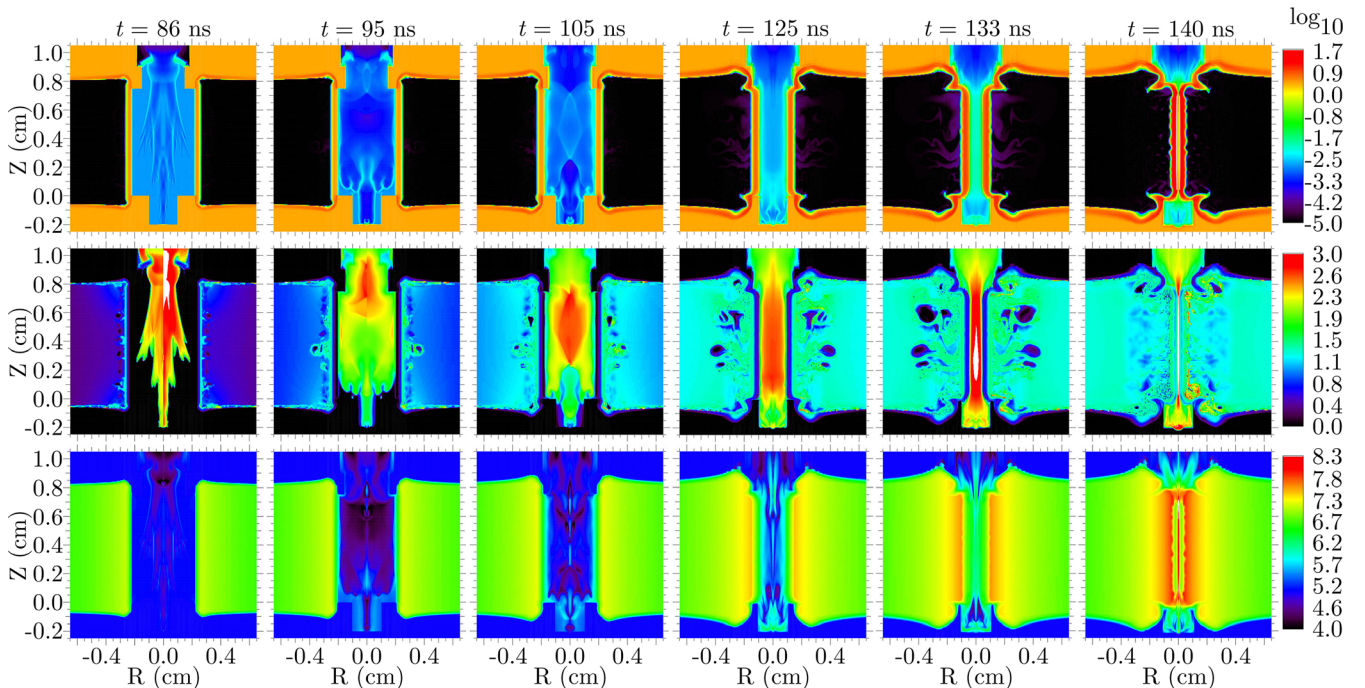


FIG. 6. Near-term integrated MagLIF density in g/cc (top), temperature in eV (middle, T_i [left] and T_e [right]), and B field magnitude in G (bottom) for several times between the laser preheat and stagnation. All plots are on a $\log_{10}$ scale.

there. Also, the wall instability grows from the liner-electrode interaction.⁴⁰ The combined result is both ends converge farther and form a magnetic bottle (or mirror) configuration at stagnation, as shown in Fig. 7, and influences confinement at late time. The liner converges until it reaches total pressure balance $P_{tot} = P_{mat} + P_{mag}$ with the fuel, where P_{mat} is the material pressure and P_{mag} is the magnetic pressure. If the B_z^0 is too large for the obtained convergence, the stagnation plasma β can fall below unity and the yield is reduced due to the implosion energy compressing the field rather than the fuel. Figure 7 reveals the majority contribution is from fuel P_{mat} at the midplane but from P_{mag} at the ends since CR varies along the liner. The low plasma pressures at the ends do not contribute much yield and the burn size $\Delta z \sim 6$ mm is shortened compared with the $L_l = 7.5$ mm.

At stagnation, the total absorbed energies in the liner and gas reach 425 kJ and 26 kJ, respectively, as illustrated in Fig. 8. Approximately 86% of the fuel energy is internal energy and the fuel mass lost by stagnation is $m_{loss} = 43\%$. The compressed B_z^f flux in the fuel is of order 100 MG, although about $\Phi_{loss} = 38\%$ of the initial flux has been lost, and the peak fields at the ends of the bottle exceed 200 MG. The magnetized fuel parameter $r_{stag}/r_L^z = 2.0$ or equivalently $B_z^f r_{stag} = 5.3 \times 10^5$ G cm. The peak B_z^f in the hot spot can be larger than B_θ in the liner (~ 70 MG) since the axial current does not diffuse all the way through the liner.

The thermonuclear fuel assembly has radius $r_g^{stag} \approx 63 \mu\text{m}$, $\rho_g \approx 0.5 \text{ g/cc}$, $\rho R_g \approx 5 \text{ mg cm}^{-2}$ and peak $T_i \approx 6.5 \text{ keV}$, resulting in 2.2 Gbar of pressure. Note the achieved peak T_i is 30% higher in the integrated calculation compared with the 1D simulation due to mass loss, extra convergence, and difference in B field evolution. The liner compression is similar to the 1D case ($\rho_l \approx 60\text{--}70 \text{ g/cc}$, $\rho R_l \approx 0.9 \text{ g cm}^{-2}$) but has some spatial variation. The primary fusion neutron yield is $Y_n^{DD} = 6.1 \times 10^{13}$ or $E_{fus} = 71 \text{ J}$, over a burn duration $t_{burn} \approx 4 \text{ ns}$ ($t_{burn}^{FWHM} = 2.1 \text{ ns}$), as also shown in Fig. 8. The primary yield and burn duration in the integrated simulation are

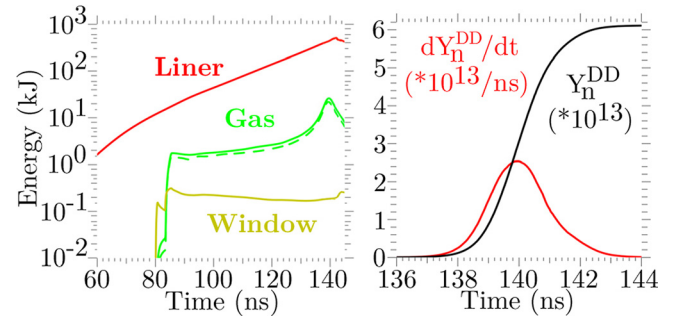


FIG. 8. On the left, energy histories for the liner (red), gas (green, internal energy is dashed), and window (yellow). On the right, primary neutron yield Y_n^{DD} (black) and production rate dY_n^{DD}/dt (red).

24% and 67% of their 1D values, respectively. The ratio of primary to secondary neutron yields has increased in 2D to $Y_n^{ratio} = 44$ from the value of 23 in 1D. The burn-weighted ion temperature that would be inferred from neutron diagnostics is $\langle T_i \rangle = 3.2 \text{ keV}$. Due to the design considerations discussed earlier, higher Z impurities are not present in the simulated stagnation column, and so the above results represent optimized performance expectations for the first near-term MagLIF experiments.

In order to generate realistic synthetic neutron spectra, the densities, temperatures, and B field profiles generated by the HYDRA code just prior to stagnation are initialized into the particle-in-cell code LSP. The fusion reactants and products, liner, and magnetic field are then evolved in r - z geometry through the burn phase using methods similar to those first reported in Refs. 41–44. The particle method is more sophisticated than the multi-group charged-particle diffusion with anisotropic conduction model used by HYDRA because the kinetic ion species obey an equation of motion resolving the cyclotron frequency and no assumptions of mean-free-path and gradient scale lengths are required. All of the relevant species are included in the model (Be , e , D , T , p , He^3 , He^4 , and n). Individual particle scattering events between all species are handled using a binary Coulomb interaction treatment, except for the high density e - Be interaction which assumes a Maxwellian distribution. Individual particle fusion reactions are modeled using a generalized binary treatment.

The density profiles for all the ions, the D ion temperature, and B field are plotted in Fig. 9 at the time of peak neutron production rate ($t = 140 \text{ ns}$). The T and He^3 particle densities peak on-axis, whereas the p density does not and the He^4 alpha particles preferentially congregate at the two ends of the magnetic bottle, where B_z is largest. Confinement of alpha particles by the $r_{stag}/r_L^z = 2.0$ value is significant, about 55% are trapped within the fuel (the rest escape to the liner boundary), in agreement with an estimate from previous work.²⁷ The primary neutron yield Y_n^{DD} is 80% of the value reported by HYDRA and the ratio $Y_n^{ratio} = 41$ instead of 44, due to differences in the transport and fusion treatments.

The primary and secondary fusion neutrons are tracked in order to produce synthetic neutron energy spectra that would be measured by time-of-flight detectors located in separate radial and axial directions on Z .³⁶ The synthetic

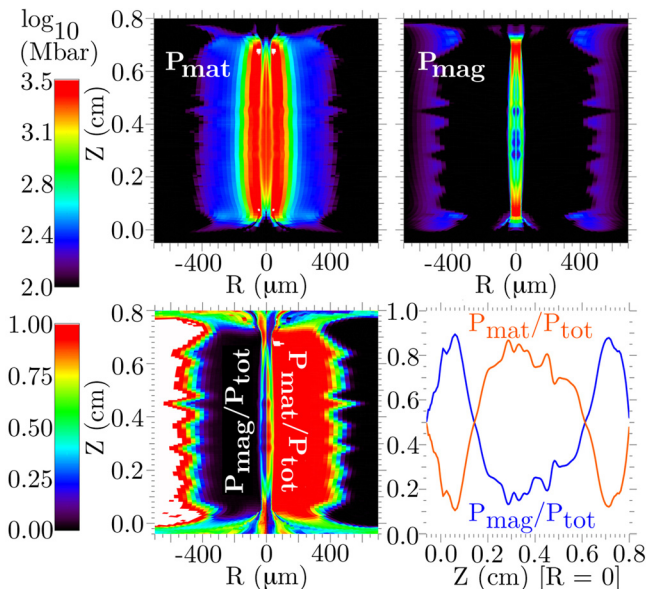


FIG. 7. Material P_{mat} and magnetic P_{mag} pressures in the magnetic bottle configuration at stagnation.

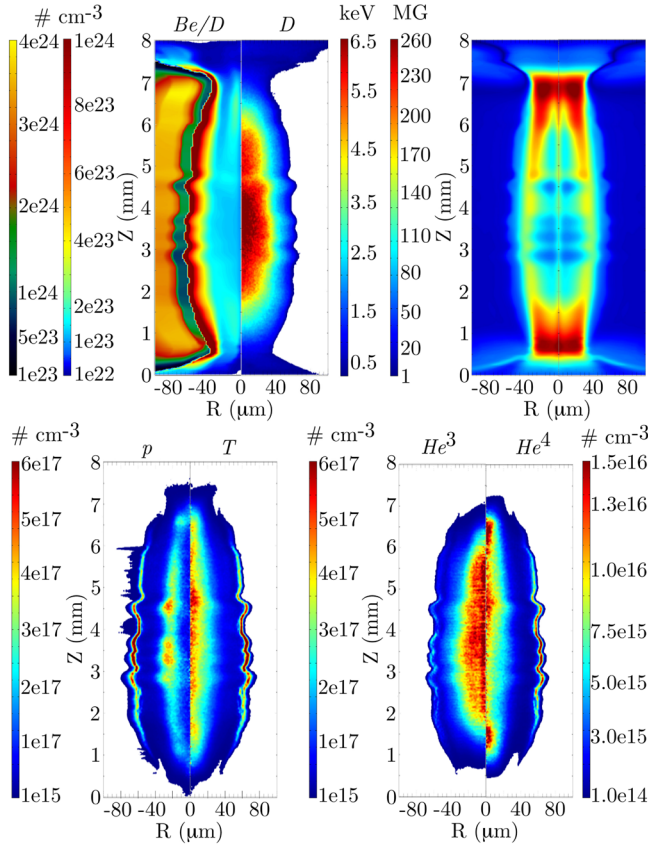


FIG. 9. Stagnation profiles from LSP simulation at time of peak burn: Be and D densities and D ion temperature (top left); B field magnitude (top right); p and T densities from DD reactions (bottom left); and He^3 and He^4 densities from DD and DT reactions, respectively (bottom right). All DD fusion products are on the same scale.

spectra are provided in Fig. 10. Isotropy in the low-energy 2.45 MeV neutrons is simulated, indicative of the neutrons' thermonuclear origin from a Maxwellian ion distribution. When n - Be scattering is treated, a low-energy tail to the otherwise Gaussian distribution is present in the radial and axial directions due to the liner ρR and ρZ , respectively. As already mentioned, the DD product T particles are magnetized and responsible for generating the He^4 and high-energy 14.1 MeV neutrons, whose spectral shape is expected to be different depending on the direction to the detector. The anisotropy and asymmetry is due to temporal-averaging of the

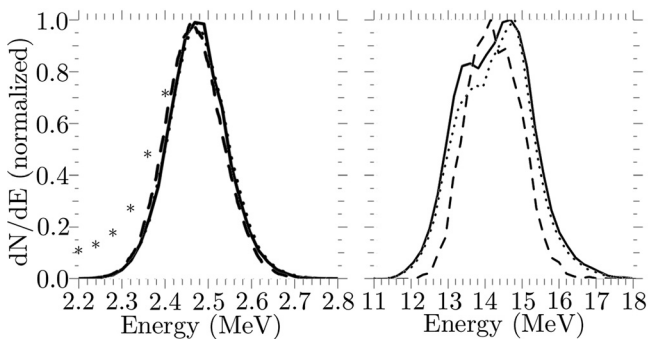


FIG. 10. Time-integrated synthetic neutron spectra generated by the LSP code, as they would be detected by neutron time-of-flight diagnostics in the radial (dashed), top axial (dotted), and bottom axial (solid) directions. A low-energy tail results from inclusion of n - Be scattering (asterisks).

spatial inhomogeneity in the compressed plasma and magnetic field profiles producing the neutrons.

B. Design variations for near-term targets

The near-term target design discussed above that uses gas density $\rho_g = 1.5$ mg/cc and liner length $L_l = 7.5$ mm is just one possible realization of MagLIF on Z. In order to evaluate trade-offs in performance due to these parameter choices, variations in ρ_g and L_l are simulated, while the liner aspect ratio and mass per unit length are held fixed to keep the implosion histories similar. Although only two seemingly innocuous parameters are altered, the ramifications for target design are complex. Increasing ρ_g increases the required window thickness dz_{win} to hold the pressure if the laser entrance hole diameter remains fixed. The increased window mass either requires more laser energy E_{las} at the same radius r_{las} to disassemble the window, and so is less efficient at coupling energy into the gas E_g^{abs} , or may be penetrated more efficiently with smaller r_{las} , but cause increased penetration depth into the gas. Increasing L_l increases the permitted penetration depth of the laser, and so reduces the optimal r_{las} , but increases the inductance of the load and so decreases the peak current I_{max} , absorbed liner energy per unit length, and implosion velocity. For these reasons, any ρ_g and L_l variations require re-optimization in order to maximize E_g^{abs} for the E_{las} used, without directly ablating any of the surrounding material.

Table I summarizes the optimized results for 12 cases wherein the integrated simulation is repeated with $\rho_g = 0.8, 1.1, 1.5$, and 2.0 mg/cc and $L_l = 5.0, 7.5$, and 10.0 mm. The densities correspond to 5–14% of the critical density for 2ω light in DD . The optimized r_{las} is found to decrease both as L_l increases with fixed ρ_g and as ρ_g increases with fixed L_l , since $\rho_g \Delta z$ increases in both situations; the range in r_{las} gives irradiance values $I_{las} = 0.3$ – 2.5×10^{14} W cm $^{-2}$. The E_g^{abs} is found to increase as L_l increases with fixed ρ_g , but does not monotonically increase as ρ_g increases with fixed L_l because of the constraint to penetrate the window without ablating other surfaces. Note the normalized E_g^{abs} cm $^{-1}$ decreases with increasing L_l for fixed ρ_g , as does the peak preheat temperature (which also decreases with increasing ρ_g).

The target performance at stagnation is quantified in Table I by the midplane liner convergence CR_{2D} , fuel mass lost m_{loss} , magnetic B_z flux lost Φ_{loss} , primary neutron yield Y_n^{DD} , primary to secondary neutron yield ratio Y_n^{ratio} , burn-averaged ion temperature $\langle T_i \rangle$, and magnetized fuel parameter r_{stag}/r_L^z . The results illustrate the m_{loss} decreases either with increasing L_l or ρ_g , since end losses are reduced whether the liner is longer or the plasma is cooler; conversely, Φ_{loss} exhibits the opposite behavior. The lowest CR_{2D} and highest fraction of yields relative to 1D are found at $L_l = 10$ mm because mass loss is minimized (the yields are 5%, 24%, and 32% of 1D for the $\rho_g = 1.5$ mg/cc cases). The Y_n^{DD} varies from 1.9 to 8.3×10^{13} because of the interplay between consequences of optimizing the preheating. The range of Y_n^{ratio} is 22–160, $\langle T_i \rangle$ is 2.2–4.4 keV, and r_{stag}/r_L^z is 1.6–2.9, or equivalently $B_z^l r_{stag}$ is 4.4 – 7.9×10^5 G cm. Independent calculations of these scenarios using the LASNEX

TABLE I. Near-term integrated MagLIF simulations for $\rho_g = 0.8, 1.1, 1.5$, and 2.0 mg/cc and $L_l = 5.0, 7.5$, and 10.0 mm .

Parameters	L_l I_{max}	5 mm 20 MA	7.5 mm 19 MA	10 mm 18 MA
$\rho_g = 0.8 \text{ mg/cc}$ $dz_{win} \approx 1.1 \mu\text{m}$	r_{las}	1100 μm	840 μm	660 μm
	E_g^{abs}	1.27 kJ	1.54 kJ	1.74 kJ
	CR_{2D}	46	42	33
	m_{loss}	73%	60%	43%
	Φ_{loss}	26%	31%	34%
	Y_n^{DD}	1.4×10^{13}	6.5×10^{13}	8.3×10^{13}
	Y_n^{ratio}	160	22	30
	$\langle T_i \rangle$	3.4 keV	4.4 keV	3.6 keV
	r_{stag}/r_L^2	2.9	2.5	1.9
$\rho_g = 1.1 \text{ mg/cc}$ $dz_{win} \approx 1.5 \mu\text{m}$	r_{las}	700 μm	530 μm	490 μm
	E_g^{abs}	1.49 kJ	1.78 kJ	1.85 kJ
	CR_{2D}	42	37	33
	m_{loss}	68%	51%	39%
	Φ_{loss}	32%	34%	37%
	Y_n^{DD}	2.0×10^{13}	7.0×10^{13}	6.5×10^{13}
	Y_n^{ratio}	43	30	50
	$\langle T_i \rangle$	3.8 keV	3.5 keV	3.0 keV
	r_{stag}/r_L^2	2.4	2.1	1.8
$\rho_g = 1.5 \text{ mg/cc}$ $dz_{win} \approx 2.0 \mu\text{m}$	r_{las}	470 μm	460 μm	440 μm
	E_g^{abs}	1.67 kJ	1.74 kJ	1.85 kJ
	CR_{2D}	40	37	33
	m_{loss}	61%	43%	35%
	Φ_{loss}	35%	38%	41%
	Y_n^{DD}	2.3×10^{13}	6.1×10^{13}	4.8×10^{13}
	Y_n^{ratio}	85	44	98
	$\langle T_i \rangle$	3.1 keV	3.2 keV	2.4 keV
	r_{stag}/r_L^2	2.2	2.0	1.7
$\rho_g = 2.0 \text{ mg/cc}$ $dz_{win} \approx 2.7 \mu\text{m}$	r_{las}	440 μm	380 μm	360 μm
	E_g^{abs}	1.52 kJ	1.62 kJ	1.66 kJ
	CR_{2D}	42	39	36
	m_{loss}	56%	35%	29%
	Φ_{loss}	38%	44%	47%
	Y_n^{DD}	3.3×10^{13}	2.5×10^{13}	1.9×10^{13}
	Y_n^{ratio}	66	110	140
	$\langle T_i \rangle$	2.9 keV	2.3 keV	2.2 keV
	r_{stag}/r_L^2	2.2	1.9	1.6

code^{9,45} recover Y_n^{DD} in Table I within an approximate factor of 2 and predict similar CR_{2D} ; they account for additional B_z flux loss from inclusion of the Nernst term, which by itself results in slightly cooler plasma with yield reductions of $\sim 10\text{--}40\%$ depending on the case.

The first experiments¹⁰ were intended to be the $\rho_g = 1.5 \text{ mg/cc}$ and $L_l = 7.5 \text{ mm}$ case, but the gas density was lowered to 0.7 mg/cc (similar to the 0.8 mg/cc case), used windows with undeformed thicknesses of $3.2 \pm 0.2 \mu\text{m}$, and $r_{las} = 200\text{--}300 \mu\text{m}$. Separate laser-only experiments⁴⁶ indicated laser coupling to the gas was significantly less than anticipated compared with the $E_g^{abs} \sim 1.5\text{--}1.7 \text{ kJ}$ in simulations using a smooth beam profile. The initial experiments measured stagnation radii $r_{stag} < 75 \mu\text{m}$ (inferred $CR \sim 40$), temperatures around 3 keV, and isotropic neutron yields up to $Y_n^{DD} = 2 \times 10^{12}$, with inferred alpha-particle

magnetization parameters around $r_{stag}/r_L^2 = 1.7$ and less than 10% Be mix.¹⁰ The primary reason for the lower-than-expected yields is currently hypothesized to be largely due to reduced laser preheat energies because of laser-plasma interaction.

C. Laser timing and coupling efficiency

An important consideration to be addressed by integrated simulation is the question of when to trigger the laser preheat relative to the accelerator. The simulation in Sec. IV A is repeated with the laser timing varied from $t = 80 \text{ ns}$ by $\pm 40 \text{ ns}$ in 5 ns steps. The optimal time balances competing deleterious effects, and occurs when the inner liner boundary begins to implode, as displayed in Fig 11. Preheating too early increases end losses, plasma cooling, and potential for mix. Preheating too late results in less effective compression. Note the yield variation away from the optimal time would be considerably worse if constant CR_{2D} were assumed for comparison, since the majority of the yield is produced at peak compression. Aside from producing yield, optimal timing also improves stability by minimizing convergence. The ZBL timing jitter is $\pm 4.5 \text{ ns}$ (2σ) and ensures preheating occurs near peak potential yield due to laser timing.

Since there likely will be un-modeled laser energy losses due to laser plasma interaction (LPI), another consideration is to understand target performance as a function of coupled laser energy. A primary concern is that simulations do not properly account for LPI within the window and gas. High-energy lasers can contain hot spots and modulations to an extent that their spatial profiles may not be well approximated as the $n = 4$ supergaussian used herein. Rather than a smooth beam with a meaningful W cm^{-2} value, which is capable of generating uniform pressure to cleanly disassemble the window and mostly heat the gas uniformly,^{47,48} laser energy may instead be concentrated in many separated regions and interact fundamentally differently with plasma than HYDRA would predict. The LPI may generate hot electrons and stimulated scattering from plasma waves and decay modes, reflect and refract laser light, and ultimately result in reduced energy coupling to the fuel. Even a smooth beam is

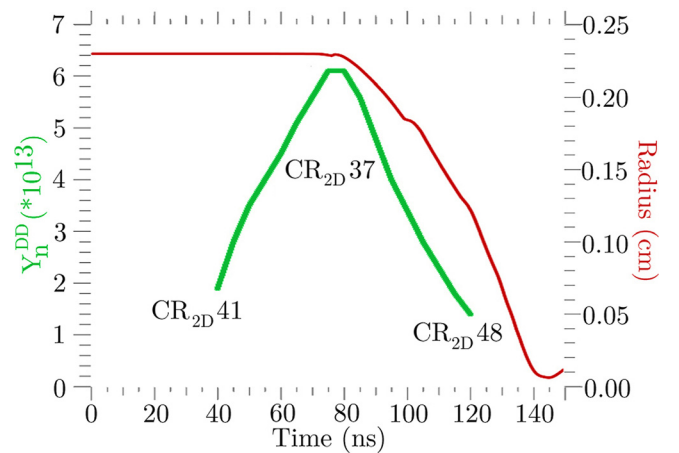


FIG. 11. Neutron yield variations (green) due to laser preheat timing. The liner trajectory is shown (red).

known to produce consequential LPI effects, such as stimulated Raman scattering and two-plasmon decay, near 25% of the critical density. The general problem of modeling an unsmooth beam may involve steep gradients and long collisional mean free paths inappropriate for local diffusion models, more advanced laser propagation than ray tracing, and kinetic treatment for the plasma and its waves. Modeling the LPI more accurately is beyond the scope of this work and the model in use.

An assessment of yield degradation due to reduced laser energy coupling to the fuel may be made by simply injecting less energy into the simulation. The simulation using $\rho_g = 1.5 \text{ mg/cc}$ is repeated with the main pulse duration reduced in steps from the original 2 ns down to 0 ns, but maintaining $P_{\text{las}} = 1 \text{ TW}$, so that E_{las} from 2.2 kJ down to 0.2 kJ (just the pre-pulse) is injected. The results are provided in Fig. 12 as Y_n^{DD} plotted versus E_g^{abs} . Reduced absorption fractions and penetration depths lead to cooler and shorter stagnation plasmas, with steadily declining yield. However, since less of the gas is heated, the fraction of mass lost by stagnation improves. When the absorbed fuel energy is only 174 J, an order of magnitude less than the intended design value, the yield falls nearly 50-fold to $Y_n^{DD} = 1.3 \times 10^{12}$, burn duration shortens to $t_{\text{burn}}^{\text{FWHM}} = 1.4 \text{ ns}$ (from 2.1 ns), $CR_{2D} = 45$ (from 37), and the burn-averaged $\langle T_i \rangle = 2.0 \text{ keV}$ (from 3.2 keV). Results from a sequence using the $\rho_g = 0.7 \text{ mg/cc}$ from experiments are also given in Fig. 12. In this case, the yield remains higher as E_g^{abs} is reduced, since the plasma is hotter ($\langle T_i \rangle$ is 50% higher on average). As Fig. 12 shows, absorbed gas energies around $E_g^{abs} \sim 100\text{--}300 \text{ J}$ in simulations are closer to the initial experimental yields $Y_n^{DD} = 0.5\text{--}2.0 \times 10^{12}$,¹⁰ and in approximate agreement with laser-only experiments showing less than roughly 500 J transmission energies through LEH windows for the parameters involved.⁴⁶ The results from laser-only experiments, as well as detailed comparison between experimental observables and post-shot degraded simulations, will be the topic of future publications.

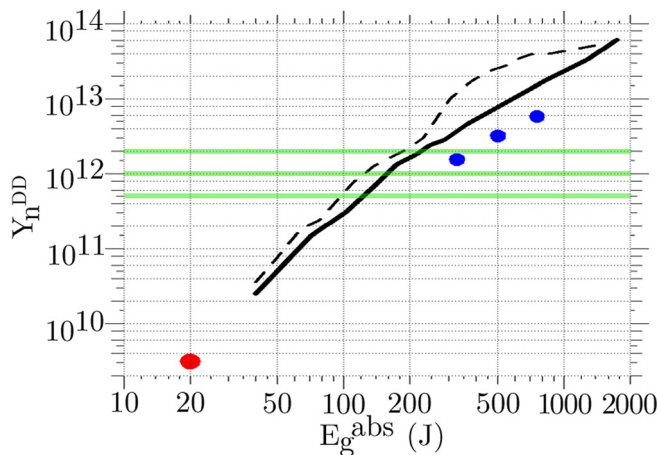


FIG. 12. Neutron yield variations due to reduced coupling of laser energy by shortening the main pulse duration for $\rho_g = 1.5 \text{ mg/cc}$ (solid) and $\rho_g = 0.7 \text{ mg/cc}$ (dashed). Experimental yields¹⁰ are denoted as green lines. Even assuming some backscatter loss and energy deposition only into the window (see text), energy can still couple to the gas and produce yield (blue dots). Without any laser preheat, $Y_n^{DD} = 3 \times 10^9$ (red dot) due to preheat from the shock generated by the current's magnetic pressure.

Even if the laser only barely penetrates the window (or not at all), the window and shocks still couple energy to the fuel, some of which can be compressed since the B_z^0 does not inhibit axial thermal conduction. To evaluate the effect of having such worst-case backscatter with LPI that prevents any direct laser coupling to the fuel, three additional simulations only deposit energy into the window. The window explosion then couples about one third of the deposited energy into the gas, and the three cases of 0%, 30%, and 60% backscatter are represented as blue dots in Fig. 12, in order of decreasing yield. Note they lie below the curves where laser energy couples directly to the fuel, and their yields can be quickly lowered if window material is mixed into the intended burn region.

When no laser energy is injected at all, the magnetic pressure generated in the liner by the current launches a shock and is the source of a small amount of preheat, resulting in the 3×10^9 neutron yield represented by the red dot in Fig. 12, consistent with liner-only flux compression experiments without laser preheat.^{10,29,31}

V. DESIGN OF FUTURE EXPERIMENTS

A. Upgraded experiment on Z

The simulations discussed in Sec. IV are used to design experiments using only capabilities that presently exist. Target performance improvement and reduction of risk associated with stably achieving large liner convergence require upgrades in the accelerator, magnetization, and preheat capabilities. Due to the significant fusion reactivity improvement, the use of DT rather than DD fuel is expected to improve yields over 50-fold by itself. The safe use of T on the Z facility is being studied.

To achieve higher currents, a charging voltage of 95 kV can be used instead of 80 kV. Current delivery can also be improved by reducing the initial inductance L_c^0 of the circuit. While Z is capable of $I_{\text{max}} = 27 \text{ MA}$ in low L_c^0 loads, MagLIF currently uses high L_c^0 transmission lines to accommodate the magnetic field coils. Increased I_{max} raises the pressure generated, the absorbed target energy (enabling more massive and/or faster liners with increased ρR), and the amount of PdV work done on the fuel. Circuit modeling suggests $I_{\text{max}} = 22 \text{ MA}$ or more may be achieved within MagLIF loads on Z using 95 kV.

To further reduce thermal conduction losses from the preheated fuel, and benefit from alpha particle self-heating at higher yields, the initial B_z^0 can be increased. Stronger B_z^0 may enable higher plasma temperatures and larger yields at smaller CR.⁹ Engineering efforts are underway to provide $B_z^0 = 30\text{--}40 \text{ T}$.

Increased laser preheat energy E_{las} implies the use of larger laser radius, higher initial gas density, and/or a longer liner to absorb the energy. In general, more preheat energy can reduce end losses (through longer L_i), enable lower velocity and aspect ratio liners (improved instability robustness), and reduce the required CR (stability risk). Planned upgrades for ZBL will increase the available energy to $E_{\text{las}} = 6 \text{ kJ}$ or more.

Simulation reveals the advantage of capability upgrades being developed for future experiments, which are closer to

the original point design.⁹ The initialization is similar to Fig. 4. The *Be* liner has aspect ratio 6, outer radius $r_l = 0.3$ cm, length $L_l = 10$ mm, mass 160 mg, and contains $\rho_g = 1.2$ mg/cc ($dz_{win} \approx 1.6$ μ m). The fuel is DT, $B_z^0 = 40$ T, and $E_{las} = 6.2$ kJ with $r_{las} = 700$ μ m. The liner trajectory and current are similar in shape to those in Fig. 2, except stagnation occurs 6 ns earlier and $I_{max} = 23.7$ MA. Preheating occurs as the implosion starts, and the gas and window absorb $E_g^{abs} = 4.7$ kJ and $E_{win} = 405$ J from it; the plasma equilibrates to an average $T \approx 600$ eV after 25 ns.

Profiles of density and temperatures are plotted in Fig. 13. Due to increased E_{las} , in this case $CR_{2D} \approx 26$, not much larger than $CR_{1D} \approx 23$. By stagnation and peak burn, the total absorbed energies for the liner and gas are 895 kJ and 68 kJ, respectively. The fuel mass and flux losses are $m_{loss} = 44\%$ and $\Phi_{loss} = 32\%$. The fuel assembly has radius $r_g^{stag} \approx 96$ μ m, $\rho_g \approx 0.3$ g/cc, $\rho R_g \approx 4$ mg cm⁻², and peak $T_i \approx 15$ keV ($\langle T_i \rangle = 8$ keV), resulting in $P_{stag} = 3.4$ Gbar. The liner compression achieves $\rho_l \approx 90$ –100 g/cc and $\rho R_l \approx 1.2$ g cm⁻². The magnetized fuel parameter $r_{stag}/r_L^2 = 6.5$, or equivalently $B_z^f r_{stag} = 1.8 \times 10^6$ G cm, and is handily within the ignition space discussed in Refs. 9 and 27. The yield is $Y_n^{DT} = 3.0 \times 10^{16}$, or $E_{fus} = 84$ kJ, over

a burn duration $t_{burn} \approx 4$ ns ($t_{burn}^{FWHM} = 2.1$ ns). The fusion energy gain relative to the total absorbed fuel energy is $G_f = 1.2$ and this achievement would represent an important scientific milestone for magneto-inertial fusion approaches, just as recent experiments achieving fuel gain exceeding unity do for traditional ICF.³⁷

B. High-gain MagLIF on a future facility

Clean 1D yields increase to 100 MJ/cm using 25 kJ of preheat energy and $I_{max} \sim 70$ MA on a future pulsed-power accelerator.^{9,49} Even though alpha particle deposition plays an increasingly important role in self-heating at higher currents, fusion gains only remain around 10 because the standard MagLIF approach volumetrically burns the low-density fuel; too much energy is invested into heating all the fuel to ignition-relevant T_i , rather than heating a small hot spot and radially propagating a burn wave into surrounding compressed fuel, as is done in traditional ICF.^{6–8}

In principle, MagLIF could achieve high gain using cryogenic solid DT fuel and substantial fuel preheat.⁴⁹ When a cryogenic DT layer is adjacent to the inner boundary of the liner, simulations indicated the gain curves rose only slightly more rapidly than their pure-gas counterparts beginning at $I_{max} \sim 30$ MA. By $I_{max} \sim 55$ MA, however, the curves for cryogenic fuel quickly separated from the volumetric curves and approached 10 GJ/cm yield with gain relative to the target $G_t = 1000$ at $I_{max} = 70$ MA.⁴⁹ An intermediate regime exists wherein the B_z field is strong enough to reduce conduction losses during the compression phase like standard MagLIF, but weak enough to not inhibit the burn wave carried by the charged alpha particles at stagnation. Ignition temperatures are realized through preheated gas compression, not the use of well-timed shocks as in traditional ICF.

An ideal 1D HYDRA simulation of a high-gain target provides an estimate to compare with more realistic integrated calculations. The *Be* liner has aspect ratio 6, outer radius $r_l = 0.488$ cm, and mass 428 mg cm⁻¹. The solid DT fuel layer has 42 mg cm⁻¹, $\rho_f = 0.25$ g/cc, and aspect ratio 5.5. The interior gas $\rho_g = 5$ mg/cc (1.7 mg cm⁻¹), which corresponds to a higher pressure than the vapor pressure of the ice layer and is 28% of the critical density for 2ω light. The $B_z^0 = 8$ T and the circuit model is the same, except the voltage is increased so $I_{max} = 70$ MA by $t = 110$ ns, as displayed in Fig. 14.

The simulation increments the internal energy of the gas to deposit an equivalent 21 kJ cm⁻¹ from $t = 65$ –97 ns (0.66 TW), in order to mock up the laser preheat. Lower preheat power helps reduce ablation and adiabat increase of the high-density fuel, where the adiabat $\alpha_f \sim P/P_{deg}$ is the fuel pressure relative to the Fermi degenerate pressure. At the end of the heating phase, $E_g^{abs} = 17$ kJ cm⁻¹ due to radiation loss and coupling to the solid fuel.

The implosion phase lasts 50 ns and achieves $v_{imp} \approx 150$ km/s, with liner $CR_{1D} \approx 24$ and fuel $CR_{1D} \approx 35$. At stagnation, the gas, fuel, and liner absorbed energies are 0.6, 1.4, and 7.7 MJ cm⁻¹, respectively ($E_t = 9.7$ MJ cm⁻¹, only 55% of which is kinetic energy). The flux loss is $\Phi_{loss} = 55\%$ and $\Phi_{loss} = 24\%$ from the gas and fuel,

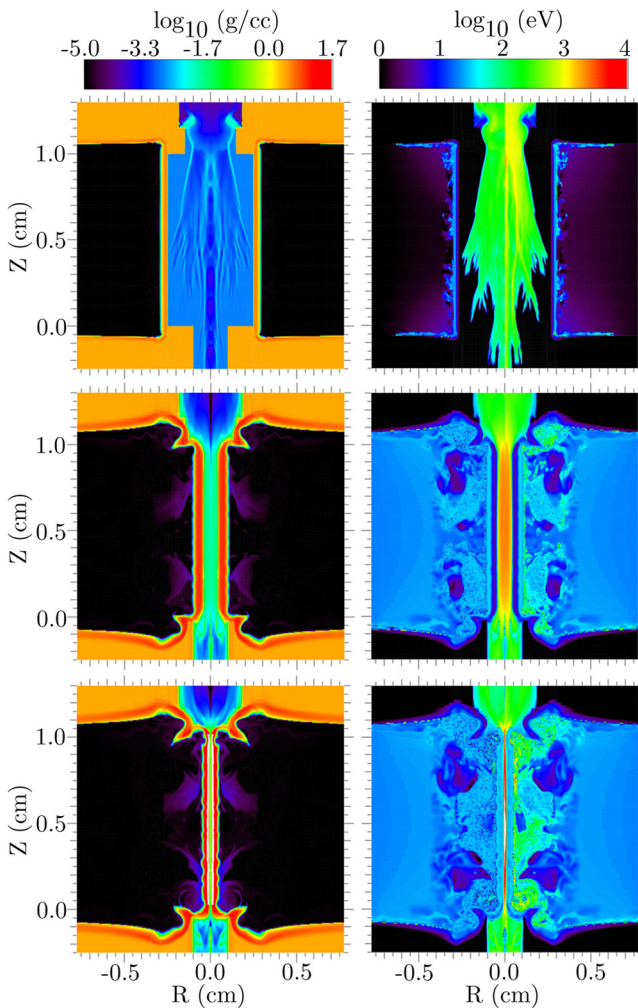


FIG. 13. Future upgraded integrated MagLIF densities (left) and temperatures (right, T_i [left] and T_e [right]) at the end of preheating 57 ns before stagnation (top), 6 ns before stagnation (middle), and stagnation (bottom).

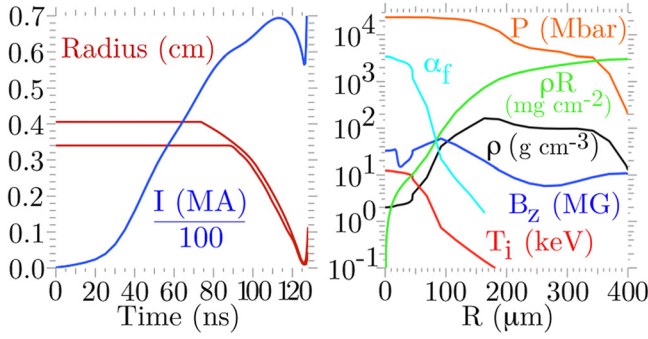


FIG. 14. High-gain MagLIF simulation in 1D. On the left, solid fuel and liner trajectories (red) and current (blue). On the right, stagnation profiles of ρ in g/cc (black), T_i in keV (red), ρR in mg cm⁻² (green), B_z in MG (blue), P in Mbar (orange), and fuel adiabat $\alpha_f \sim P/P_{deg}$ (cyan).

respectively, but note that its loss at late time is not deleterious like it is for standard MagLIF, since the burn wave must overcome the conduction inhibition of the compressed flux.

Stagnation profiles are shown in Fig. 14, the time corresponds to peak no-burn absorbed energy just before alpha deposition becomes important (and $E_{fus} \approx E_g + E_f$). The hot spot (gas) and fuel densities are $\rho_{hs} \approx 2\text{--}45$ g/cc and $\rho_f \approx 45\text{--}160$ g/cc, with $r_{hs}^{stag} \approx 95$ μm and $r_f^{stag} \approx 169$ μm . The areal densities in the hot spot, fuel, and liner are $\rho R_{hs} \approx 0.1$ g cm⁻², $\rho R_f \approx 0.8$ g cm⁻², and $\rho R_l \approx 3.1$ g cm⁻², with peak $T_i \approx 13$ keV before ignition. The magnetization parameters are $r_{stag}/r_L^z = 2.2$ and $B_z^f r_{stag} = 5.9 \times 10^5$ G cm.

Although ρR_{hs} and ρR_f are not much less than required for standard ICF, some benefit is gained via magnetization since $P_{stag} = 25$ Gbar just prior to significant alpha deposition and ignition, compared with the hundreds of Gbar required for NIF capsules.^{6,8} The yield is $Y_n^{DT} = 2.8 \times 10^{21}$ cm⁻¹, or $E_{fus} = 7.9$ GJ cm⁻¹, which is gain $G_f = 4000$ relative to the fuel and $G_t = 810$ relative to the target. The burned fuel fraction is 53%, quite high due to the tamping effect of the liner.¹⁷ Although the $t_{burn} \approx 0.4$ ns during ignition, the target produces neutrons with measurable and rapidly increasing rate for the entirety of the compression phase after being preheated.

The integrated high-gain simulation initialization is illustrated in Fig. 15. The liner $L_l = 10$ mm, $E_{las} = 25$ kJ with $r_{las} = 1$ mm, and the current, masses, and trajectory are identical to the 1D case; the 20% increase in E_{las} compensates for expected losses. The LEH remains the same, except sub-micron dz_{win} is allowed due to low pressure in the cryogenic target. At the end of preheating, the gas $E_g^{abs} = 23$ kJ, of which 80% is internal energy.

Profiles of the density and temperatures are shown in Fig. 16 at three times, which correspond to T equilibration after the heating phase, just before ignition, and just after ignition. The plasma equilibrates to $T \approx 700$ eV around 20 ns after the laser is off, and $T_i \sim 10\text{--}15$ keV in the column prior to ignition. The gas mass loss is $m_{loss} = 30\%$. Stagnation values for r_{stag} , ρR , and P_{stag} similar to those in Fig. 14 are reached just before ignition, and the plentiful preheat has allowed for $CR_{2D} \approx CR_{1D}$. Unlike the preheated gaseous fuel in

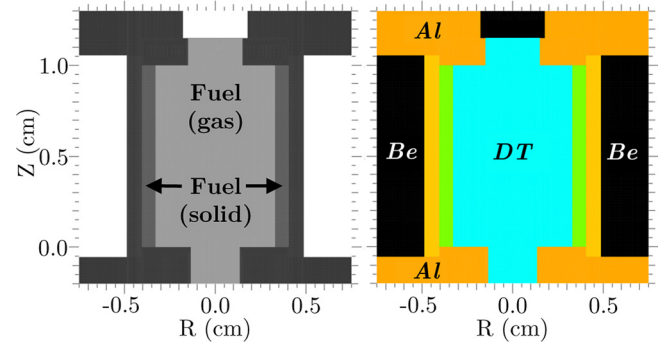


FIG. 15. Integrated high-gain MagLIF initialization using cryogenic fuel.

standard MagLIF, the cryogenic main fuel layer is too dense and cold to move away from the imploding region, implying a majority of the 1D yield may be recovered during ignition. During the radial burn wave propagation, T_i of $\sim 50\text{--}80$ keV ($\langle T_i \rangle = 34$ keV) and P of 100s of Gbar are generated. The fusion yield is $Y_n^{DT} = 2.1 \times 10^{21}$, or $E_{fus} = 5.8$ GJ. Therefore, the integrated 2D gains are $G_f = 2900$ and $G_t = 600$ relative to the fuel and target, or 73% of the 1D result and a 39% burn-up fraction.

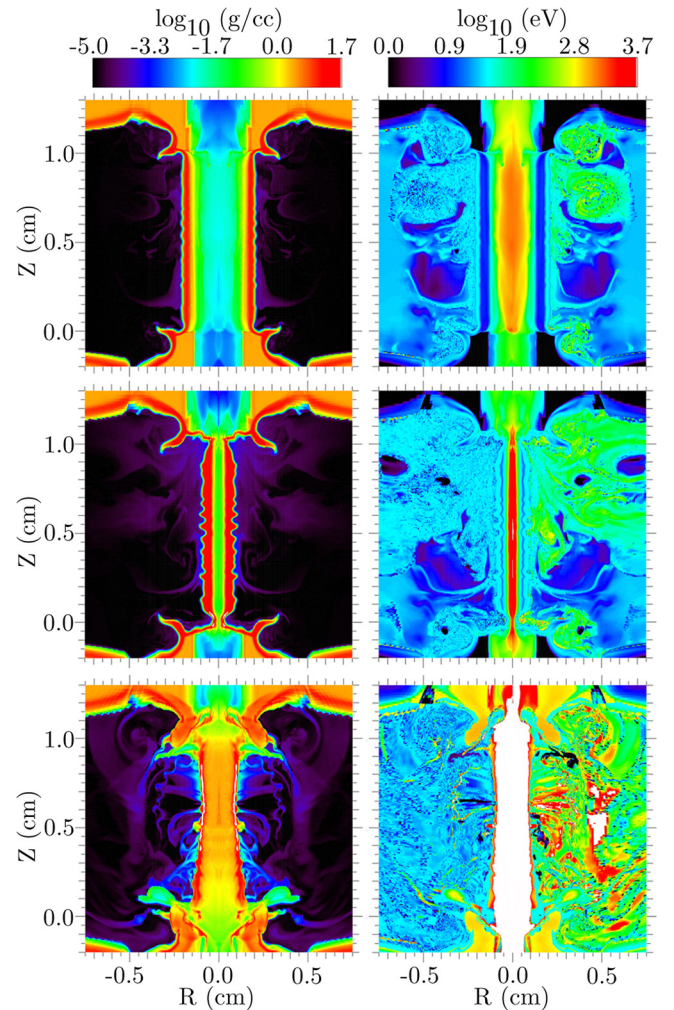


FIG. 16. Integrated high-gain MagLIF densities (left) and temperatures (right, T_i [left] and T_e [right]) at 10 ns before stagnation (top), 2 ns before stagnation (middle), and 1 ns after ignition (bottom).

C. Alternative fuel preheating method

It is important to note the gas density used in the high-gain target of Sec. V B ($\rho_g = 5$ mg/cc) is substantially higher than the equilibrium gas density ($\rho_g = 0.3$ mg/cc) of the solid cryogenic DT layer near its triple point. In addition, higher gas densities are closer to the critical density of the laser and so LPI effects may become more important in this regime. An active area of current MagLIF research is to investigate alternative methods for realizing the necessary density and preheated equilibration temperature conditions for the magnetized fuel plasma prior to compression.

One such promising method utilizes the cryogenic cooling capability already available in Z experiments. Rather than pressurizing the target (≤ 30 atm), losing available pre-heat energy to disassembly of a thick window, and then predictably depositing energy in the magnetized fuel without ablating surrounding materials, the beam dump in the bottom electrode (cathode) is replaced with a reservoir of the cryogenic high-density fusion fuel. The laser is focused onto the reservoir and ablates fuel, filling the target with hot magnetized plasma. Since the target is cryogenic, neither the very thin window nor the equilibrium gas density directly absorb much of the laser energy, and LPI risk is reduced. The absorption site is restricted to the critical density surface ablated near the reservoir. The laser radius and temporal power profile require optimization. Smaller r_{las} creates hotter fuel that can escape out the LEH and may over-ablate high-density fuel into the intended hot spot (preventing its heating), while larger r_{las} may not produce hot enough plasma. The pre-pulse may generate a longer gradient scale length for enhanced absorption. Flexibility is increased in terms of absorbing large E_{las} within $L_t = 10$ mm target lengths.

To illustrate the potential of the method, the integrated high-gain simulation is repeated with the reservoir and equilibrium gas density filling the target. The $E_{las} = 25$ kJ beam is focused upon the reservoir with $r_{las} = 50$ μ m. Profiles of density and temperatures resulting from deposition into the reservoir are shown in Fig. 17. The B_z flux compression evolves differently but is not less effective. By the plasma equilibration time, the central gas ρ_g and T are within a factor of 2 of the $\rho_g = 5$ mg/cc simulation values (compare the Fig. 17 bottom plots to the Fig. 16 top plots). Bremsstrahlung losses and shock production are altered, reducing the ablation of main fuel into the hot spot and changing its adiabat history. The case ignites similarly.

The alternative method may be tested on Z using current capabilities. The near-term integrated design simulation in Sec. IV A is repeated except with sub-micron dz_{win} , a reservoir of cryogenic DD in the beam dump, and equilibrium ρ_g (without the solid layer adjacent to the liner). The same $E_{las} = 2.2$ kJ is used, but with $r_{las} = 375$ μ m focused onto the reservoir, delivering 390 J and 1.38 kJ to the gas and fuel, respectively. Peak values of $\rho_g \approx 0.5$ g/cc and $T_i \approx 6$ –7 keV with $\langle T_i \rangle = 4.5$ keV are achieved. The yield is $Y_n^{DD} = 2.7 \times 10^{13}$ with $Y_n^{ratio} = 27$.

VI. SUMMARY AND DISCUSSION

A parameter study was conducted to optimize the target design for initial MagLIF experiments. Those integrated calculations provide realistic design requirements for neutron-

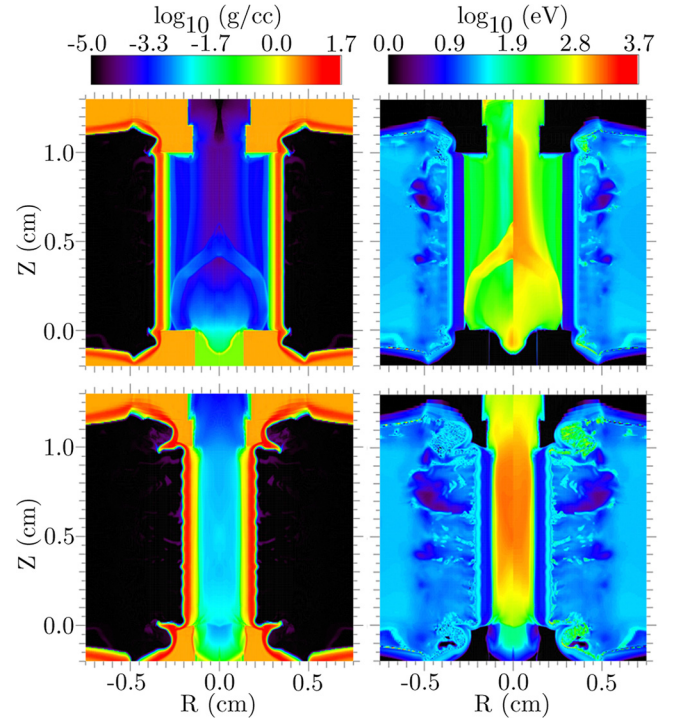


FIG. 17. Integrated alternative-preheat high-gain MagLIF densities (left) and temperatures (right, T_i [left] and T_e [right]) at the end of the laser pulse 28 ns before stagnation (top) and 10 ns before stagnation (bottom). Compare the bottom plots with the top plots in Fig. 16.

producing experiments as well as optimized performance expectations. Observables include the current, implosion time, ρ_g , ρR_g , ρR_t , T_e , convergence ratio, burn time, and the radial and axial extent of the burn region (from x-ray diagnostics); neutron diagnostics measure primary Y_n^{DD} and secondary Y_n^{DT} neutron yields, burn-averaged ion temperature $\langle T_i \rangle$, inferred r_{stag}/r_L^2 , and neutron spectra in separate directions. Detailed comparison between the observables and post-shot simulations using reduced absorbed gas energies from the laser preheat will be the topic of a future publication, as will simulation benchmarking efforts to ongoing focused experiments involving flux compression (liner and B_z only) and fuel preheating (laser and B_z only).

Integrated MagLIF simulations using HYDRA predict neutron yields $Y_n^{DD} = 2$ – 8×10^{13} are possible on Z using available capabilities for near-term experiments; simulated yields are somewhat lower when the Nernst term and more accurate burn models are used. Initial experiments¹⁰ have generated $Y_n^{DD} = 0.5$ – 2.0×10^{12} yields with expected 3 keV temperatures, $CR \sim 40$, low mix, isotropic DD yields and spectra with anisotropic DT spectra, primary to secondary neutron yield ratios $Y_n^{ratio} = 40$ –80 indicative of magnetization, and are consistent with the observables from integrated simulations similar to those presented in this paper using low pre-heat energies coupled to the fuel. Upgraded capabilities in future experiments may allow for fusion yields on the order of the absorbed gas energy. High-gain designs with substantial preheat energy on a future accelerator may enable a path to ignition at relatively low stagnation pressures.

Table II summarizes the simulated performance of integrated MagLIF targets with present parameters on Z,

TABLE II. Integrated MagLIF scaling performance.

Parameters	Present on Z	Upgrades on Z	New machine
L_l	7.5 mm	10 mm	10 mm
E_{las}	2.2 kJ	6.2 kJ	25 kJ
B_z^0	10 T	40 T	8 T
ρ_g	1.5 mg/cc	1.2 mg/cc	5.0 mg/cc
Fuel	DD	DD or DT	DT
I_{max}	19 MA	24 MA	70 MA
$\langle T_i \rangle$	3.2 keV	8 keV	34 keV
ρR_l	0.9 g cm ⁻²	1.2 g cm ⁻²	3.1 g cm ⁻²
P^{stag}	2.2 Gbar	3.4 Gbar	25 Gbar
CR_{2D}	37	26	24
Y_n	6.1×10^{13}	3.7×10^{14} (DD) 3.0×10^{16} (DT)	2.1×10^{21}
E_{fus}	71 J	0.43 kJ (DD) 84.5 kJ (DT)	5.8 GJ
G_f	0.003	0.006 (DD) 1.2 (DT)	2900

upgraded parameters on Z, and a future machine. The table parameters are liner length L_l , laser energy E_{las} , initial field B_z^0 , fuel gas density ρ_g , fuel species, peak current I_{max} , burn-averaged ion temperature $\langle T_i \rangle$, liner areal density ρR_l , peak pressure at stagnation P^{stag} , required convergence CR_{2D} , neutron yield Y_n , fusion energy E_{fus} , and gain relative to the fuel G_f . Longer L_l is beneficial for reducing fuel end losses, higher B_z^0 is preferred for reducing thermal conduction losses, and larger E_{las} reduces the required convergence.

Aside from increasing the available E_{las} and B_z^0 , possibilities to improve MagLIF performance are numerous and the subject of active research. Liner stability in high-convergence implosions is affected by magneto-RT growth seeded by surface roughness and electrothermal instability, and thick dielectric coatings are known to suppress the seed amplitude by tamping the early time blowoff.⁵⁰ Enabling the stable implosion of higher aspect ratio liners would allow for increased v_{imp} and result in reduced end losses and higher yields from fixed preheat energy. Attempts to diagnose dynamic mix between the hot fuel and other materials will be the subject of focused and integrated experiments and modeling. Alternative methods to preheat the fuel, such as a cryogenic reservoir of fuel in the beam dump, may avoid potential laser coupling and LPI issues while also enabling more efficient absorption of preheat energy. Reduction of the initial load inductance L_C^0 , whether through optimized transmission lines or alternative B_z generation methods, would increase the delivered current and target absorbed energy. Quasi-isentropic compression of the liner (or DT fuel layer, in the case of high-gain) using a shaped current pulse is expected to have benefits, such as increased current and energy delivery to the target, higher liner v_{imp} , ρ_l , and ρR_l , better fuel compression and confinement, and higher gain from lower adiabat fuel assemblies (high-gain targets). Pulse shaping may allow the future use of smaller and more economical long-pulse accelerators, wherein the $L \frac{dI}{dt}$ voltage and convolute losses are reduced. Circuit modeling predicts $I_{max} = 27$ MA may be achieved in long-pulse (300 ns) mode

on Z, and integrated simulations show promise that MagLIF can also produce fusion yield on those longer timescales, although may suffer increased plasma cooling and lifetime issues.

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